

The hidden structure of synesthetic associations: a mixture model framework

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Synesthetic associations are often described as idiosyncratic, yet large datasets reveal consistent patterns. Many past studies discovered “rules” that govern these associations, but these studies examine each rule in isolation, so it is difficult to compare them quantitatively or understand how they jointly influence synesthetic associations. I propose a mixture model framework in which synesthetic associations arise from latent factor-specific probability distributions whose relative contributions can be estimated from behavioral data. Here, I give three examples of how this framework enables direct comparison of RFs, accounts for group- and individual differences, and demonstrates that observed variation in synesthetic associations is often meaningful structure rather than idiosyncratic noise.

Study 1: Touch-Color

The Data: Associations for 20 tactile surfaces

- Data from Root et al. (2025)⁷ and unpublished master’s theses^{8,9}
- Participants (N=45) use a color picker¹⁰ to report Touch→Color associations
- Predictor data: Touch→Emotion (N=12), Touch→Object (N=14)
Object→Color (N=54), Emotion→Color (N=53)

The Rules: Touch-Color influenced by [Emotions, Objects]

- Emotion: Sticky is disgusting + disgusting is green = sticky is green
- Object: Rough is like sandpaper + sandpaper is brown = rough is brown

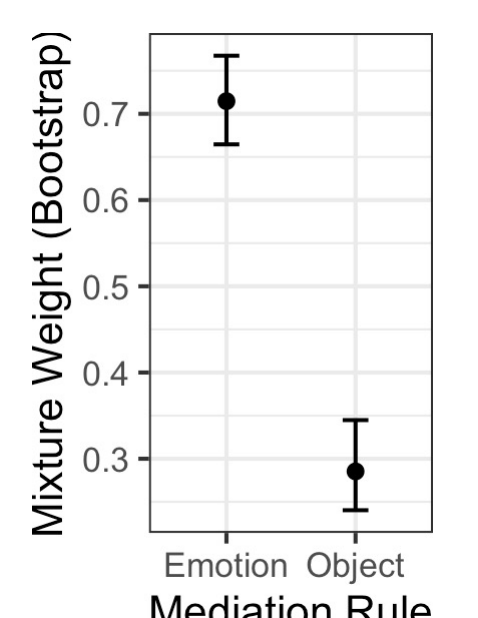
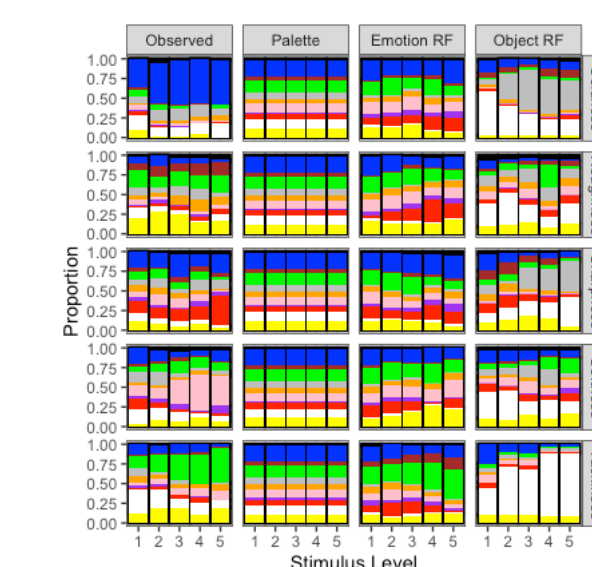
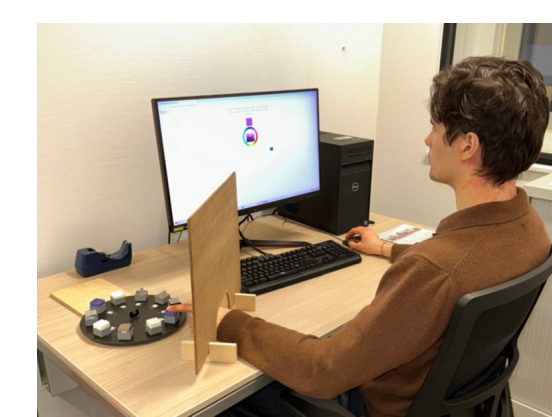
The Question: Is one rule stronger than the other?

The Model: Categorical mixture model of two distributions

- $\theta_1 = \sum_{emo} p(emo|stim)p(color|emo)$, $\theta_2 = \sum_{obj} p(obj|stim)p(color|obj)$
- $\hat{p}(color|stim) = \sum_k \alpha_k \theta_k$, mixture weights fit via maximum likelihood, bootstrap CIs

The Finding: Emotion is a better predictor than object

- $z_{boot} = 5.73, p < 0.001$

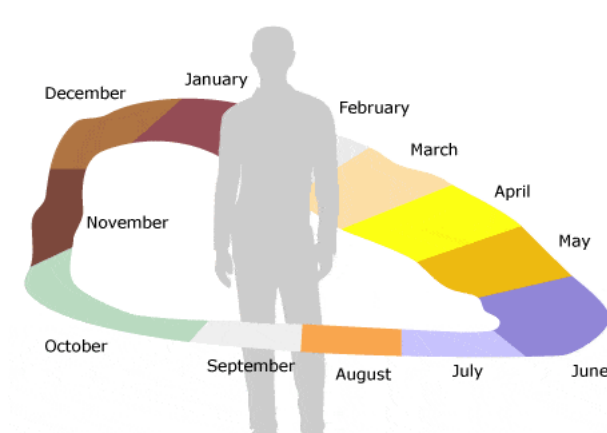


What is synesthesia?

- A stimulus in one modality elicits an automatic, consistent, counterfactual experience in another modality¹
- E.g., for grapheme-color synesthetes, letters have a color
- Synesthetes’ specific associations (e.g., *which* letter is *which* color) were once believed to be idiosyncratic²
- Large-N studies have shown that there are many patterns³
- Many different influences, e.g., for grapheme-color synesthesia: shape⁴, pronunciation⁵, and semantics⁶
- To date, these influences were studied in isolation, making the effect size of any individual influence seem small



SYNESTHESIA
0123456789



Study 2: Grapheme-Color

The Data: Letter-color associations from 20 languages

- Open data from Cross Language Synesthesia Consortium (40+ authors)¹¹
- N>2000 synesthetes, >150000 reported color associations
- Participants use a color picker¹⁰ to indicate the color of each letter

The Rule: Similarly-pronounced letters get similar colors

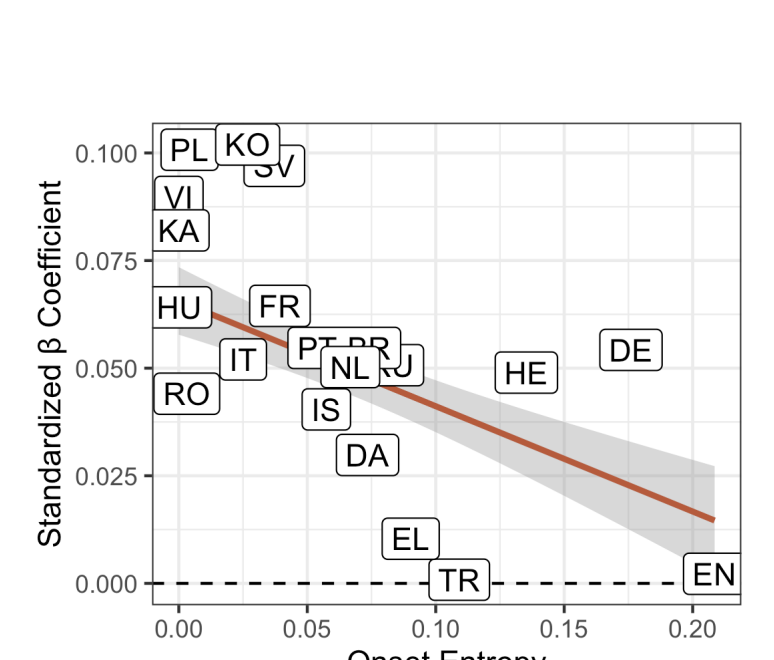
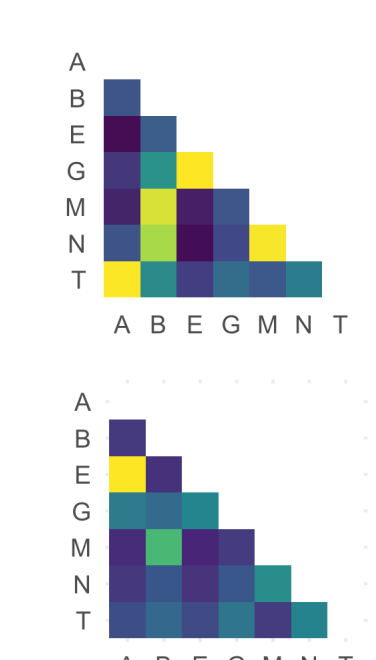
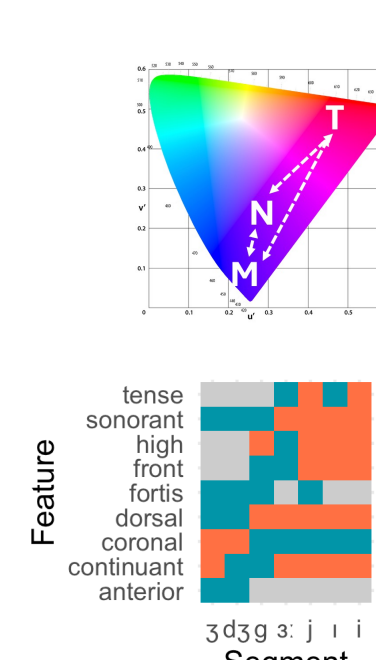
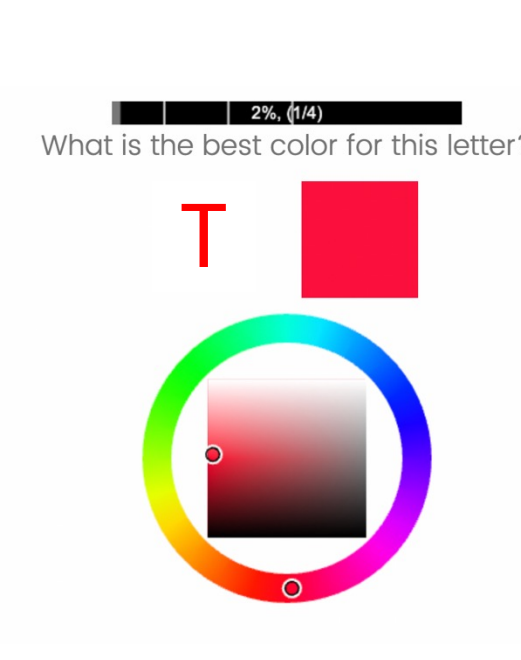
The Question: Is the rule stronger in transparent orthographies?

The Model: Hierarchical model with depth as Level 2 predictor

- $c = \text{color (CIELuv)}$, $\theta = \text{pronunciation (SNC)}$, $H = \text{transparency (entropy)}$, $\ell = \text{language}$
- $\Delta_{c_{ij\ell}} \sim \text{Norm}(\beta_0 + \beta_1 \Delta_{\theta_{ij\ell}}, \sigma)$, $\beta_1 \sim \text{Norm}(\gamma_0 + \gamma_1 H_{\ell}, \tau)$, γ_1 fit via Maximum Likelihood

The Finding: The effect size depends on orthographic depth

- $\chi^2(1) = 96.2, p < 0.001$



A Mixture Model Framework

- A “rule” k is a set of predictions θ_k : the observations expected if all observations were explained by rule k
- Observed data is a mixture: $\sum_k \alpha_k \theta_k$ where $\sum_k \alpha_k = 1$
- For MLE: $\alpha_k = \text{softmax}(\beta_k)$ and optimize on β_k
- (Optional) can optimize a predictor of β_k rather than optimizing β_k itself: explain why rule strength differs
- Likelihood Ratio Test on β_k : does it explain variance?
- Bootstrap α_k CIs: is one rule stronger than another?
- Pseudo- R^2 : how much variance is explained overall?

Study 3: Music-Color

The Data: Sound-color associations

- Open data from Ward et al. (2025)¹²
- N=68 synesthetes, 24 tone stimuli: 3 timbres (instruments) x 8 pitches
- Participants use a color picker to indicate the color of each sound

The Rules: Color influenced by...

- Timbre: e.g., flute sounds are blue, violin sounds are orange
- Pitch: e.g., Ab is purple, Eb is brown, regardless of the octave
- Frequency: pitches with higher fundamental frequency are brighter colors

The Question: Are there subtypes of music-color synesthesia?

The Model: Mixture model of three RDM-like predictors

- (θ_1) same timbre, same color; (θ_2) same pitch, same color; (θ_3) higher freq., brighter color
- $\Delta_{c_{ij}} = \sum_k \alpha_k \theta_k$, mixture weights fit via per-subject maximum likelihood

The Finding: Pitch-color is a (rare) distinct form of synesthesia, whereas timbre- and frequency-color can co-occur

